Differential Geometry: midsem exam

Aditya Padiyar

September 2026

Definition 1. *A submanifold S is called totally geodesic if for every $p \in S$ and $v \in T_pS$ we have range $\gamma_v \subseteq S$*

Definition 2. *We say that $X \in \mathcal{X}(M)$ is compatible with S if $X_{|S} \in \mathcal{X}(S)$. This space is $\mathcal{X}_S(M)$*

Claim 1. *If $X, Y \in \mathcal{X}_S(M)$ then $[X,Y]_{|S} = [X_{|S}, Y_{|S}]_S$*

1 Second fundamental form

Fix a submanifold S. Let ∇ and $\tilde{\nabla}$ denote the Levi-Civita connections on M and S respectively.

Definition 3. *Define $\mathcal{F} \in B(\mathcal{X}_S(M), C^\infty(S, \mathbb{R}^k))$ by*

$$X, Y \to (\nabla_X Y) - \tilde{\nabla}_X Y$$

Claim 2. *range $\mathcal{F} \subseteq \mathcal{Y}(S)$*

Proof. Observe that $[\tilde{\nabla}_X Y](q) = \tilde{\pi}([\nabla_X Y](q))$ □

Claim 3. *$\mathcal{F}$ is symmetric*

Claim 4. *$\mathcal{F}(fX, Y) = f\mathcal{F}(X,Y)$ for any $f \in C^\infty(M)$*

Claim 5. *If $X(p) = 0$ then $\mathcal{F}(X,Y)(p) = 0$*

Proof. Choose a tangential frame $X_i \in \mathcal{X}(U)$ on some neighborhood U. We have $X = \sum f_i X_i$ with $f_i(p) = 0$. Now use the bilinearity with respect to smooth functions. □

Claim 6. *Suppose $v \in T_pS$. Then there is a neighborhood U of p and $X \in \mathcal{X}_{S\cap U}(U)$ such that $X(p) = v$*

Proof. Let $\phi : \Omega \to U$ be a local parametrization compatible with S, with inverse ψ. Let $\varphi : W \to S \cap U$ is the associated local parametrization of S. Now $v = (D\varphi)_0 y$. Define $X : U \to \mathbb{R}^k$ by $X(q) = (D\psi)_q^{-1}(y, 0)$ □

Claim 7. *Suppose $v_1, v_2 \in T_pS$ and $X_i, Y_i \in \mathcal{X}_S(M)$ are such that*

$$X_i(p) = Y_i(p) = v_i$$

Then $\mathcal{F}(X_1, X_2)(p) = \mathcal{F}(Y_1, Y_2)(p)$

Proof. We have $\mathcal{F}(X_1, X_2) - \mathcal{F}(Y_1, Y_2) = \mathcal{F}(X_1 - Y_1, X_2) - \mathcal{F}(Y_1, X_2 - Y_2)$. But $X_1 - Y_1$ and $X_2 - Y_2$ vanish at p □

Definition 4. *We have $Q_p \in B(T_pS, N_pS)$ defined by $(v, w) \to \mathcal{F}(X, Y)(p)$*

Claim 8. *Q_p is symmetric*

Claim 9. *Suppose γ is a smooth curve in S defined on an interval I and V is a smooth vector field along γ satisfying $V(t) \in T_{\gamma(t)}S$. Then we have*

$$(cov\ V)(t) = (\widetilde{cov}\ V)(t) + Q_{\gamma(t)}(\gamma'(t), V(t))$$

where cov and $\widetilde{cov}$ denote the covariant derivatives along γ in M and S respectively.

Proof. Consider the following

- Choose t_0 and a compatible local parametrization $\phi : \Omega \to U$
- The associated local parametrization of S is $\varphi : W \to S \cap U$
- Let $\gamma(t_0) = p$ and $T : \mathbb{R}^k \to N_pS$ be the orthogonal projection
- $V(t) = V_i(t)\varphi_i(\gamma(t))$ and $\gamma(t) = \varphi(x(t))$
- $Q_p(\gamma'(t_0), V(t_0)) = \dot{x_i}(t_0)V_j(t_0)Q_p(\varphi_i(p), \varphi_j(p))$
- $\dot{V}(t_0) = \dot{x_i}(t_0)V_j(t_0)(D_{\varphi_i}\varphi_j)(p) + \dot{V_j}(t_0)\varphi_j(p)$
- $(\text{cov} - \widetilde{\text{cov}})V(t_0) = T(\dot{V}(t_0)) = \dot{x_i}(t_0)V_j(t_0)T(D_{\varphi_i}\varphi_j)(p)$
- Now use the definition of Q_p. The required vector field extensions are ϕ_i and ϕ_j respectively

□

Claim 10. *If S is totally geodesic then $Q_p = 0$ for all $p \in S$*

2 Fixed points

Suppose $\mathcal{A} \subseteq \mathrm{Iso}(M)$, where $\mathrm{Iso}(M)$ denotes the group of Riemannian isometries on M. Define

$$C = \{q \in M : q \text{ is a fixed point of every } A \in \mathcal{A}\}$$

Claim 11. *C is closed in M*

Claim 12. *C is a submanifold of M*

For $q \in C$ define the subspace W_q of T_qM to be the fixed points of DF_q

Claim 13. *Suppose $q \in C$ and $v \in \mathcal{E}_q \cap W_q$. Then $\exp_q(v) \in C$*

Claim 14. *Every connected component of C is a submanifold of M*

Claim 15. *We have $T_qC = W_q$*

Suppose now that C is a connected one dimensional submanifold of M and $\sigma : \mathbb{R} \to C$ is a diffeomorphism

Claim 16. *We have $(cov\ \sigma')(t) \in W_{\sigma(s)}$ where the covariant derivative is in M*

Claim 17. *If $|\sigma'(t)| = 1$ then σ is a geodesic.*

3 Uniformly normal neighborhood

Suppose W is a uniformly normal neighborhood.

Claim 18. *Suppose $v_i \in T_pM$ are unit vectors and $q_i = \exp_p(c_i v_i)$ are distinct points in W for $i = 1, 2, \ldots, n$. Define $f : W \to [0, \infty)$ by $f(x) = \sum_{i=1}^n d_M(q_i, x)$ and suppose f attains its minimum at p. Then we have $\sum_{i=1}^n v_i = 0$*

4 Removal of a point

Suppose M is connected and $p \in M$. Define $N = M \setminus \{p\}$

Claim 19. *N is a connected submanifold*

Claim 20. *For all $q_1, q_2 \in N$ we have $d_N(q_1, q_2) = d_M(q_1, q_2)$*